16

$$v = \sqrt{\frac{a}{b}}$$

$a = 6.43$ correct to 2 decimal places.
$b = 5.514$ correct to 3 decimal places.

By considering bounds, work out the value of v to a suitable degree of accuracy.
Give a reason for your answer.

$6.425 \leq a < 6.435$

1 mark
Half-unit bounds

$5.5135 \leq b < 5.5145$

Lower bound of $v = \sqrt{(6.425 \div 5.5145)}$

1 mark
Use a bound and a b bound in formula

$= 1.079402688...$

Upper bound of $v = \sqrt{(6.435 \div 5.5135)}$

1 mark
Small ÷ large for LB; large ÷ small for UB

$= 1.080340323...$

1 mark
Both correct bounds

Both bounds round to 1.08 to 2 decimal places.

So $v = 1.08$ to 2 d.p.; every possible value rounds to this.

1 mark
1.08 with reason

(Total for Question 16 is 5 marks)